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KAPASA MAKASA UNIVERSITY

SCHOOL OF APPLIED SCIENCES

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

ACCESS CONTROL AND INTRUSION DETECTION QUIZ TEST – 2025

TIME: 30 MINUTES

Instructions: *Circle the correct answer*

1. Which of the following protocols is commonly used to collect information about CPU utilization and band width from network devices and automatically send the administrator an email if a predefined threshold is exceeded?

(A) HTTP (B) SNMP (C) DHCP (D) SMTP

2. You have set up an Intrusion detection system (IDS) and suddenly the IDS identifies an activity as an attack but the activity is acceptable behaviour. The state, in this case, is known as:

(A) False-positive (B) False-negative (C) Non-credentialed scans (D) Credentialed scans.

3. During a vulnerability scan, you found a serious SQL injection vulnerability in one of your online shop that has 10,000 daily visitors. The shop provides 99.999% availability to customers so it can't be taken offline to fix the SQL injection vulnerability. Which of the following solutions should you recommend to the shop administrators until the shop can be remediated?

(A) Port security (B) WAF (C) Honeypot (D) Sandboxing

4. The developers of your company for efficiency purposes they need to implement a solution to deploy new code releases automatically into the production environment. Which of the following software development method the developers should implement?

(A) Continuous integration (B) Continuous monitoring (C) Continuous delivery (D) Continuous deployment

5. Which of the following process is designed to trigger automatic code integration in the main code base in stead of developing in isolation and then integrating them at the end of the development cycle?

(A) Continuous deployment (B) Continuous integration (C) Continuous monitoring (D) Continuous delivery

6. Which of the following sources of information would provide you with the most accurate information for you to use in determining which network service has been stopped working on the company's network?

(A) Event logs (B) Firewall logs (C) Syslog (D) Flow analysis

7. You are conducting a log review scan to detect the users who have been locked out from their accounts in or der to proceed to the recovery stage. Which of the following sources of information would provide you with the most accurate information for you to use in determining who has been locked out or not?

(A) Event logs (B) Firewall logs (C) Syslog (D) Flow analysis

8. You have been tasked to implement a solution to allow users from untrusted networks to access the Web Server and the Mail Server while keeping the private network secured and inaccessible. Which of the following do you need to implement to complete the task?

(A) IDS (B) IPS (C) DMZ (D) VPN

9. You have been tasked to improve the wired net work security of the company's network by limiting the number of MAC addresses on a given port. Packets that have a match ing MAC address should be considered secure packets and should be forwarded; all other packets (unsecured packets) should be restricted. Which of the following features should you implement to complete the task?

(A) Port security (B) Network access control (C) Sinkholing (D) Sandboxing

10. A new social media app was launched recently that accepts all users worldwide by default. A cybersecurity analyst after reviewing the firewall logs detects that users from specific countries have conducted a cyber attack on the app. Which of the following access control method should the ana lyst recommend to the developers?

(A) Sandboxing (B) Whitelisting (C) Port security (D) Blacklisting

11. Which of the following options is a dictionary that provides definitions for publicly disclosed cybersecurity vulnerabilities and exposures?

(A) Log aggregation (B) Security Orchestration, Automation, and Response

(C) Sentiment analysis (D) Common Vulnerabilities and Exposures

12. A developer is building a new corporate application that needs to be accessed only by selective employees based on their IP addresses. Which of the following access control method should the developer implement?

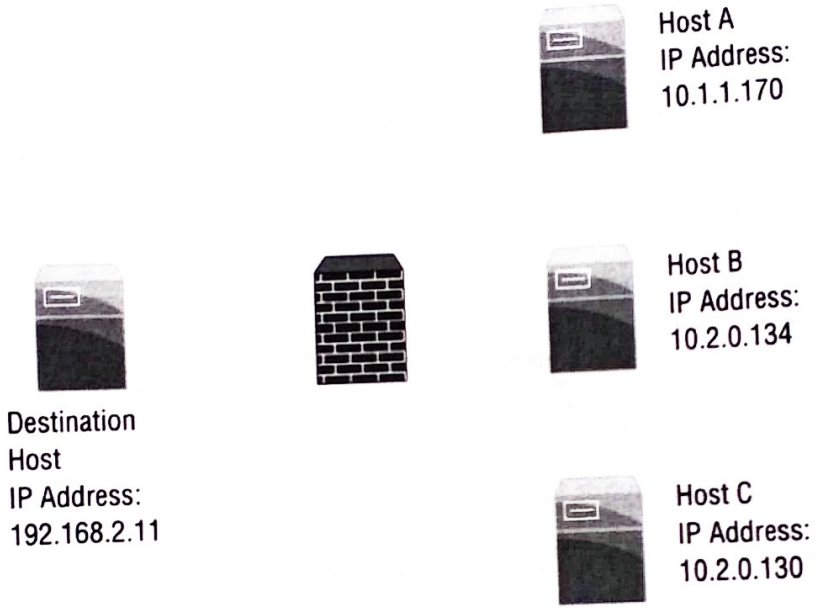
(A) Blacklisting (B) Port security (C) Whitelisting (D) Sandboxing

13. Which of the following methods should you use to deal with the huge volume of new threats seen daily?

(A) Intrusion detection system (B) Network access control

(C) Heuristics analysis (D) Intrusion prevention system

14. Cameron needs to set up a Linux iptables-based firewall ruleset to prevent access from hosts A and B, while allowing SMTP traffic from host C. Which set of the following commands will accomplish this?



A. # iptables -I INPUT 2 -s 10.1.1.170 -j DROP # iptables -I INPUT 2 -s 10.2.0.0/24 --dport 25 -j DROP # iptables -I INPUT 2 -s 10.2.0.130 --dport 25 -j ALLOW

B. # iptables -I INPUT 2 -s 10.1.1.170 -j DROP # iptables -I INPUT 2 -s 10.2.0.0.134 -j DROP # iptables -I INPUT 2 -s 10.2.0.130 --dport 25 -j ALLOW

C. # iptables -I INPUT 2 -s 10.1.1.170 -j ALLOW # iptables -I INPUT 2 -s 10.2.0.0.134 -j ALLOW # iptables -I INPUT 2 -s 10.2.0.130 --dport 25 -j DROP

D. # iptables -I INPUT 2 -s 10.1.1.170 -j DROP # iptables -I INPUT 2 -s 10.2.0.0.134 -j DROP # iptables -I INPUT 2 -s 10.2.0.130 -j ALLOW

15. Which of the following is the primary function of an Intrusion Detection System (IDS)?

(A)To actively block malicious traffic (B) To monitor network traffic and alert on suspicious activity

(C) To encrypt all network traffic

(D) To automatically patch vulnerabilities

16. How does an Intrusion Prevention System (IPS) differ from an IDS?

(A) An IPS only monitors network traffic, while an IDS can both monitor and block malicious traffic.

(B) An IPS can actively block malicious traffic, while an IDS can only generate alerts.

(C) An IDS can encrypt network traffic, while an IPS cannot.

(D) An IPS can automatically patch vulnerabilities, while an IDS cannot.

17. What is a key advantage of anomaly-based IDS?

(A)It is very effective at detecting known attacks. (B) It can detect novel or zero-day attacks. (C) It generates very few false positives. (D) It is easy to configure and maintain.

18. What is a false negative in the context of an IDS?

(A) An IDS incorrectly identifies legitimate traffic as malicious.

(B) An IDS fails to detect malicious traffic and allows it to pass.

(C) An IDS generates too many alerts, overwhelming security personnel.

(D) An IDS is unable to monitor network traffic.

19. What is a padded cell in the context of intrusion detection?

(A) A type of network firewall. (B) A decoy system used to lure attackers. (C) A system for automatically patching vulnerabilities. (D) A device that encrypts network traffic.

20. When a NIDS is pulling all traffic from the network so that it can be processed, it must be run in what mode?

(A)Traffic collector (B)Monitor (C)Promiscuous (D)Analysis